

Applications of First-Order and Stochastic Differential Equations in Yield Curve Modeling

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Abstract

This project explores the mathematical framework underlying interest rate dynamics, transitioning from deterministic models to stochastic processes. Accurately modeling the yield curve is essential for pricing fixed-income securities and assessing economic conditions. However, traditional first-order ordinary differential equations (ODEs) fail to capture the inherent, unpredictable volatility of financial markets. To address this limitation, this paper examines Stochastic Differential Equations (SDEs), with a primary focus on the Vasicek interest rate model. By incorporating both a deterministic, mean-reverting drift and a stochastic diffusion term driven by Brownian motion, the Vasicek model provides a realistic representation of short-rate movements over time. The project details the theoretical formulation of these differential equations and implements computational simulations to generate prospective interest rate paths. Ultimately, the study demonstrates how numerical methods can be used to simulate these stochastic paths, providing a foundational tool for yield curve analysis.

1 Motivation

The fundamental bedrock of modern financial engineering and quantitative analysis lies in the accurate pricing of fixed-income securities and the management of interest rate risk. At the heart of these tasks is the yield curve, a dynamic entity that reflects the broader market's collective expectation of future macroeconomic conditions. Accurately modeling the evolution of the yield curve is not merely a theoretical exercise; it is a critical requirement for algorithmic trading, derivatives pricing, and robust portfolio hedging strategies.

Traditionally, deterministic models utilizing first-order ordinary differential equations (ODEs) have been employed to describe continuous compounding and growth. While mathematically elegant, these foundational models operate under the assumption of a frictionless, predictable environment. They inherently fail to account for the empirical reality of financial markets: interest rates are subject to continuous, unpredictable shocks driven by macroeconomic data releases, central bank policy shifts, and volatile investor sentiment.

To bridge the gap between theoretical models and market reality, it is imperative to introduce stochasticity. By transitioning from standard ODEs to

Stochastic Differential Equations (SDEs), we can mathematically decompose interest rate movements into a predictable, deterministic drift component and a volatile, random diffusion component. Frameworks such as the Vasicek and Cox-Ingersoll-Ross (CIR) models provide the necessary mathematical machinery to capture vital market phenomena, such as mean reversion, while explicitly pricing in market noise.

The primary motivation for this project is to explore this critical transition from deterministic to stochastic modeling. By dissecting the theory behind these differential equations and computationally simulating their paths, this study aims to demonstrate how advanced statistical data science techniques can be applied to complex financial systems. Ultimately, this provides a more rigorous, realistic, and highly applicable framework for yield curve analysis.

2 Introduction

Interest rates are among the most consequential variables in all of modern finance. They govern the cost of borrowing, the pricing of debt instruments, the valuation of derivatives, and the broader allocation of capital across an economy. At the heart of fixed-income analysis lies a deceptively simple object: the **yield curve**, defined as the relationship between the yield (annualized return) of a set of bonds and their respective times to maturity, holding all other attributes—issuer, currency, and credit quality—constant.

The Yield Curve and Its Significance

Formally, let $T > 0$ denote a maturity horizon and let $P(t, T)$ denote the price at current time t of a zero-coupon bond that pays one unit of currency at time T . The continuously compounded spot rate, or *yield*, for that maturity is defined as:

$$R(t, T) = -\frac{\ln P(t, T)}{T - t} \quad (1)$$

The yield curve is then the mapping $T \mapsto R(t, T)$ for a fixed t . In practice, this curve is inferred from observable market prices of Treasury securities or overnight index swaps. Its shape—whether upward-sloping (*normal*), flat, or inverted—carries direct implications for monetary policy expectations,

recession forecasting, and the relative attractiveness of short- versus long-duration assets [1].

Why a Stochastic Model Is Necessary

A naive approach to interest rate modeling might treat the short-term rate $r(t)$ as a deterministic function of time—for instance, as the solution to a simple first-order ordinary differential equation (ODE). While pedagogically useful as a starting point, such a framework is fundamentally inadequate for the realities of financial markets. Interest rates fluctuate continuously in response to macroeconomic data releases, central bank policy shifts, and the collective behavior of market participants—all of which introduce genuine, irreducible uncertainty. This uncertainty cannot be captured by any deterministic trajectory.

The natural mathematical language for modeling such randomness is the theory of **stochastic differential equations (SDEs)**, which augments a deterministic differential equation with a noise term driven by a Wiener process (standard Brownian motion). SDEs allow us to characterize not just a single predicted path for $r(t)$, but the entire probability distribution of future rate trajectories—a far richer and more honest description of reality.

Scope and Structure of This Paper

This project constructs a complete, self-contained pipeline that begins with elementary calculus and terminates in a calibrated computational simulation of the yield curve. The progression is deliberate:

1. **Section 3** develops the mathematical framework in three stages: deterministic exponential growth via first-order ODEs (Section 3.1), the introduction of Brownian motion as a model for market noise (Section 3.2), and the derivation of the **Vasicek model** SDE as the primary object of study (Section 3.3).
2. **Section 4** describes the computational methodology, including the Euler–Maruyama discretization scheme and the rationale behind the chosen parameter values.

3. **Section 5** presents and interprets the simulation results: the ensemble of simulated short-rate paths and the implied yield curves constructed from them.
4. **Sections 6–8** summarize the findings, critically assess the known limitations of the Vasicek model—most notably its allowance of negative interest rates—and identify the Cox–Ingersoll–Ross (CIR) model as the immediate direction for future work.

The central thesis of this paper is that stochastic differential equations represent the minimal mathematical structure required to produce financially meaningful interest rate models, and that even the simplest such model—the Vasicek model—yields rich, interpretable, and computationally tractable dynamics that illuminate the essential mechanics of yield curve behavior.

3 Mathematical Framework

The development of a stochastic interest rate model proceeds in three conceptual stages. We begin with the simplest possible model—a deterministic first-order ODE governing exponential growth—and systematically identify its failures as a description of real markets. We then introduce the probabilistic machinery of Brownian motion to handle market noise, and finally assemble both ingredients into the Vasicek stochastic differential equation. Each stage is mathematically self-contained and directly motivates the next.

3.1 Deterministic Growth (First-Order ODEs)

The Constant-Rate Model

The most elementary model of an interest-bearing account supposes that the instantaneous rate of change of value $V(t)$ is proportional to the value itself. Letting $r > 0$ be a fixed, constant interest rate, this assumption yields the first-order linear ODE:

$$\frac{dV}{dt} = r V(t), \quad V(0) = V_0 \tag{2}$$

This is separable. Dividing both sides by $V(t)$ and integrating from 0 to t :

$$\int_0^t \frac{1}{V(s)} \frac{dV}{ds} ds = \int_0^t r ds \implies \ln V(t) - \ln V_0 = rt \tag{3}$$

Exponentiating both sides gives the classical closed-form solution:

$$\boxed{V(t) = V_0 e^{rt}} \quad (4)$$

Under this model, a zero-coupon bond paying one unit of currency at maturity T , discounted at constant rate r , has present value:

$$P(0, T) = e^{-rT} \quad (5)$$

and the yield curve $T \mapsto R(0, T)$ is trivially flat:

$$R(0, T) = -\frac{\ln P(0, T)}{T} = r \quad \text{for all } T > 0 \quad (6)$$

Allowing a Time-Varying Rate

A mild generalization replaces the constant r with a deterministic function $r(t)$, giving the non-autonomous ODE:

$$\frac{dV}{dt} = r(t) V(t), \quad V(0) = V_0 \quad (7)$$

which has the exact solution $V(t) = V_0 \exp\left(\int_0^t r(s) ds\right)$. The corresponding bond price becomes:

$$P(0, T) = \exp\left(-\int_0^T r(t) dt\right) \quad (8)$$

This can produce non-trivial yield curve shapes if $r(t)$ is prescribed analytically. However, it remains a *deterministic* model: once $r(t)$ is fixed, every future rate is known with certainty. This is the fundamental deficiency. Markets do not offer such certainty; on any given day, the short-term rate can move unpredictably in response to policy announcements, inflation data, or shifts in risk appetite. A deterministic framework has no mechanism to represent this uncertainty, and therefore cannot replicate the empirical distribution of yield curve shapes observed across time.

3.2 Introducing Stochasticity

Brownian Motion

To model continuous-time randomness rigorously, we require a stochastic process with properties that match the qualitative behavior of market noise:

continuity of paths, independence of non-overlapping increments, and normally distributed changes. These properties are axiomatized in the definition of a **standard Brownian motion** (also called a **Wiener process**).

Definition 1 (Standard Brownian Motion) *A stochastic process $\{W(t)\}_{t \geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a standard Brownian motion if it satisfies the following four conditions:*

1. $W(0) = 0$ almost surely.
2. **Independent increments:** For any $0 \leq s < t \leq u < v$, the increment $W(v) - W(u)$ is independent of $W(t) - W(s)$.
3. **Stationary Gaussian increments:** For any $0 \leq s < t$,

$$W(t) - W(s) \sim \mathcal{N}(0, t - s) \quad (9)$$

4. **Continuous paths:** The map $t \mapsto W(t)$ is continuous almost surely.

A critical consequence of property (3) is that Brownian increments over a small interval $[t, t + dt]$ satisfy:

$$dW(t) \sim \mathcal{N}(0, dt) \quad (10)$$

equivalently written $dW(t) = \sqrt{dt} Z$ where $Z \sim \mathcal{N}(0, 1)$. This $\sqrt{dt}$ scaling—arising from the variance being proportional to elapsed time—is the source of the characteristic roughness of Brownian paths and, algebraically, of the correction term in Itô’s lemma.

Stochastic Differential Equations

A general one-dimensional **stochastic differential equation** (SDE) takes the form:

$$dr(t) = \mu(t, r(t)) dt + \sigma(t, r(t)) dW(t) \quad (11)$$

where:

- $\mu(t, r)$ is the **drift coefficient**, governing the deterministic tendency of the process;
- $\sigma(t, r)$ is the **diffusion coefficient** (volatility), scaling the magnitude of random fluctuations;

- $dW(t)$ is the Brownian increment, supplying the randomness.

The expression (11) is understood in integral form: for any $t > 0$,

$$r(t) = r(0) + \int_0^t \mu(s, r(s)) ds + \int_0^t \sigma(s, r(s)) dW(s) \quad (12)$$

The first integral is an ordinary Riemann integral; the second is an **Itô stochastic integral**, defined as a mean-square limit of adapted Riemann sums. The key property of the Itô integral that distinguishes it from ordinary calculus is captured by **Itô's isometry**:

$$\mathbb{E} \left[\left(\int_0^t \sigma(s) dW(s) \right)^2 \right] = \mathbb{E} \left[\int_0^t \sigma(s)^2 ds \right] \quad (13)$$

and by the informal multiplication rule $dW(t)^2 = dt$, which lies at the heart of Itô's lemma and distinguishes stochastic calculus from its classical counterpart.

3.3 The Vasicek Model

Model Specification

Proposed by Oldřich Vašíček in 1977 [2], the Vasicek model is the canonical single-factor short-rate model. It specifies the instantaneous risk-free rate $r(t)$ as the solution to the following linear SDE:

$$\boxed{dr(t) = \kappa(\theta - r(t)) dt + \sigma dW(t)} \quad (14)$$

with parameters:

Parameter	Domain	Interpretation
κ	$\kappa > 0$	Speed of mean reversion
θ	$\theta \in \mathbb{R}$	Long-run mean (equilibrium level)
σ	$\sigma > 0$	Instantaneous volatility
$r(0)$	$r(0) \in \mathbb{R}$	Initial short rate

Economic Interpretation of the Drift

The drift term $\kappa(\theta - r(t))$ encodes **Ornstein–Uhlenbeck mean reversion**. Its mechanics are intuitive:

- If $r(t) < \theta$: the drift is *positive*, pushing the rate upward toward equilibrium.
- If $r(t) > \theta$: the drift is *negative*, pulling the rate downward toward equilibrium.
- If $r(t) = \theta$: the drift vanishes; only the Brownian noise term $\sigma dW(t)$ drives the rate.

The parameter κ controls how aggressively the process reverts: a large κ implies rapid return to θ after a shock; a small κ implies sluggish reversion and persistent deviations. This behavior is economically well-motivated: central bank policy acts as a gravitational force pulling short rates toward a target level, which is precisely what the Vasicek drift term models.

Exact Solution via the Integrating Factor Method

Because the SDE (14) is *linear* in $r(t)$, it admits a closed-form solution. Rewrite (14) as:

$$dr(t) + \kappa r(t) dt = \kappa \theta dt + \sigma dW(t) \quad (15)$$

Multiply through by the integrating factor $e^{\kappa t}$ and recognize the left-hand side as an exact differential:

$$d[e^{\kappa t} r(t)] = e^{\kappa t} (\kappa \theta dt + \sigma dW(t)) \quad (16)$$

Integrating both sides from 0 to t :

$$e^{\kappa t} r(t) - r(0) = \kappa \theta \int_0^t e^{\kappa s} ds + \sigma \int_0^t e^{\kappa s} dW(s) \quad (17)$$

Evaluating the deterministic integral and multiplying through by $e^{-\kappa t}$ yields the **exact solution**:

$$\boxed{r(t) = \theta + (r(0) - \theta) e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW(s)} \quad (18)$$

Distributional Properties

Since (18) is an affine function of a Gaussian stochastic integral, $r(t)$ is Gaussian for all $t \geq 0$. Its mean and variance follow directly from the Itô isometry (13):

$$\mathbb{E}[r(t)] = \theta + (r(0) - \theta)e^{-\kappa t} \quad (19)$$

$$\text{Var}[r(t)] = \sigma^2 \int_0^t e^{-2\kappa(t-s)} ds = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t}) \quad (20)$$

As $t \rightarrow \infty$, the process converges in distribution to a stationary Gaussian:

$$r(t) \xrightarrow{d} \mathcal{N}\left(\theta, \frac{\sigma^2}{2\kappa}\right) \quad (21)$$

Affine Bond Pricing Formula

A key strength of the Vasicek model is that it belongs to the **affine term structure** class, which allows zero-coupon bond prices to be computed in closed form. Under the risk-neutral measure $\mathbb{Q}$, the price at time t of a zero-coupon bond maturing at T is:

$$P(t, T) = A(t, T) \exp(-B(t, T) r(t)) \quad (22)$$

where, setting $\tau = T - t$:

$$B(t, T) = \frac{1 - e^{-\kappa\tau}}{\kappa} \quad (23)$$

$$A(t, T) = \exp\left[\left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B(t, T) - \tau) - \frac{\sigma^2}{4\kappa} B(t, T)^2\right] \quad (24)$$

The implied yield is then:

$$R(t, T) = -\frac{\ln P(t, T)}{\tau} = \frac{B(t, T) r(t) - \ln A(t, T)}{\tau} \quad (25)$$

Equations (22)–(25) are what we will evaluate computationally in Section 5 to construct the implied yield curve from simulated short-rate paths.

4 Computational Methodology

The closed-form solution derived in equation (18) is analytically exact, but in practice a simulation-based approach is both more flexible and more representative of the methodology used for models that do *not* admit closed-form solutions—such as the Cox–Ingersoll–Ross model discussed in Section 8. This section describes the numerical discretization scheme used to simulate sample paths of the Vasicek process, justifies the choice of that scheme, and documents the parameter values selected for all experiments reported in Section 5.

4.1 Simulation Approach: The Euler–Maruyama Scheme

Why Numerical Discretization Is Required in General

An SDE of the form $dr(t) = \mu(r, t) dt + \sigma(r, t) dW(t)$ defines $r(t)$ as a continuous-time stochastic process. Computers, however, operate in discrete time: they cannot evaluate the Itô integral $\int_0^t \sigma(r, s) dW(s)$ directly. Instead, we partition the time horizon $[0, T]$ into N equal subintervals of width $\Delta t = T/N$, and approximate the continuous dynamics by a recurrence relation evaluated at the discrete grid points $t_n = n \Delta t$ for $n = 0, 1, \dots, N$.

Even though the Vasicek model *does* admit the exact solution (18), we deliberately employ a numerical scheme here for two reasons. First, it provides a direct computational interpretation of the SDE (14)—each step of the recurrence mirrors the drift-plus-noise structure of the equation term by term. Second, it prepares a methodological template that transfers unchanged to models with no closed form, making the simulation code immediately extensible.

The Euler–Maruyama Discretization

The **Euler–Maruyama (EM) method** is the stochastic analogue of the classical forward Euler method for ODEs. Applied to the Vasicek SDE (14), it produces the following update rule:

$$r_{n+1} = r_n + \underbrace{\kappa(\theta - r_n) \Delta t}_{\text{drift increment}} + \underbrace{\sigma \sqrt{\Delta t} Z_{n+1}}_{\text{diffusion increment}} \quad (26)$$

where $\{Z_n\}_{n=1}^N$ is a sequence of independent standard normal random variables, $Z_n \sim \mathcal{N}(0, 1)$, and the factor $\sqrt{\Delta t}$ arises from the distributional

property of Brownian increments established in equation (10): $\Delta W_n = W(t_{n+1}) - W(t_n) \sim \mathcal{N}(0, \Delta t) \stackrel{d}{=} \sqrt{\Delta t} Z_n$.

The recurrence (26) is applied iteratively starting from the initial condition $r_0 = r(0)$, generating a single discrete sample path $\{r_0, r_1, \dots, r_N\}$. Repeating this procedure M independent times—each with a freshly drawn independent sequence $\{Z_n\}$ —produces an ensemble of M paths, from which Monte Carlo estimates of any functional of $r(t)$ can be computed.

Convergence and Step-Size Selection

The Euler–Maruyama method has **strong convergence order** $1/2$ and **weak convergence order** 1 [3]. Concretely, this means:

$$\text{Strong:} \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} |r(t) - r^{\Delta t}(t)| \right] = \mathcal{O}((\Delta t)^{1/2}) \quad (27)$$

$$\text{Weak:} \quad |\mathbb{E}[f(r^{\Delta t}(T))] - \mathbb{E}[f(r(T))]| = \mathcal{O}(\Delta t) \quad (28)$$

for smooth test functions f . Strong convergence controls pathwise accuracy—the proximity of individual simulated paths to true paths—while weak convergence governs the accuracy of distributional quantities such as expected bond prices, which is the object of ultimate interest here. Both orders improve as $\Delta t \rightarrow 0$.

In our simulation we set $\Delta t = 1/252$, corresponding to one trading day in a 252-trading-day year. At this resolution the discretization error in expected bond prices is of order $\mathcal{O}(1/252) \approx 4 \times 10^{-3}$, which is negligible relative to the parameter uncertainty inherent in any calibrated short-rate model. A convergence diagnostic confirming the stability of our results under halving of Δt is reported in Section 5.

Python Implementation Structure

The simulation is implemented in Python using `NumPy` for vectorized random number generation and array operations, and `Matplotlib` for visualization. The core logic is structured as follows:

1. **Parameter initialization:** Set $\kappa, \theta, \sigma, r_0, T, N$, and M (documented in Section 4.2).

2. **Random matrix generation:** Draw an $M \times N$ matrix of independent standard normals, $\mathbf{Z} \in \mathbb{R}^{M \times N}$, in a single vectorized call to `numpy.random.standard_normal`.
3. **Path simulation:** Apply the recurrence (26) column-by-column across all M paths simultaneously, avoiding any Python-level loop over paths. Only the time-step loop (over N steps) is explicit.
4. **Bond price computation:** For each maturity τ on a discrete grid, apply the affine formulae (23)–(24) to the terminal simulated rates to obtain a distribution of bond prices, then average across paths to obtain the Monte Carlo estimate $\hat{P}(0, \tau)$.
5. **Yield curve construction:** Convert $\hat{P}(0, \tau)$ to yields via $\hat{R}(0, \tau) = -\ln \hat{P}(0, \tau)/\tau$ for each τ in the maturity grid.

The complete, annotated source code is provided in the supplementary appendix.

4.2 Parameter Selection

Simulation Grid Parameters

The temporal and ensemble parameters defining the simulation grid are:

Symbol	Description	Value	Units
T	Time horizon	5	Years
N	Number of time steps	1260	Steps (252×5)
Δt	Step size	$1/252$	Years (1 trading day)
M	Number of simulated paths	1000	Paths

A horizon of five years captures the short-to-medium segment of the yield curve, which is most sensitive to monetary policy and where the mean-reversion dynamics of the Vasicek model are most visibly operative. An ensemble of $M = 1000$ paths provides Monte Carlo standard errors of order $1/\sqrt{1000} \approx 3.2\%$ on expected quantities, which is adequate for the illustrative purposes of this study.

Model Parameters

The Vasicek model parameters are:

Symbol	Description	Value	Rationale
r_0	Initial short rate	0.03	Approximate post-2010 Fed Funds rate
κ	Mean-reversion speed	0.30	Moderate reversion; half-life ≈ 2.3 yr
θ	Long-run mean	0.05	Historical average US short rate
σ	Instantaneous volatility	0.02	Consistent with low-rate-regime estimates

Each parameter choice is justified individually below.

Initial rate $r_0 = 0.03$. An initial rate of 3% is representative of the US federal funds rate in the post-global-financial-crisis era and places the process below its long-run mean, so the simulations will visibly demonstrate the upward drift induced by the mean-reversion mechanism.

Mean-reversion speed $\kappa = 0.30$. The **half-life** of a deviation from θ under the Vasicek model is the time $t_{1/2}$ at which the expected gap $\mathbb{E}[r(t)] - \theta$ has halved. From equation (19):

$$|\mathbb{E}[r(t_{1/2})] - \theta| = \frac{1}{2} |r_0 - \theta| \implies e^{-\kappa t_{1/2}} = \frac{1}{2} \implies t_{1/2} = \frac{\ln 2}{\kappa} \quad (29)$$

With $\kappa = 0.30$, this gives $t_{1/2} = \ln 2 / 0.30 \approx 2.31$ years—a moderate reversion speed consistent with empirical estimates from US Treasury data [4].

Long-run mean $\theta = 0.05$. A long-run equilibrium of 5% reflects the approximate average of the US federal funds rate over the period 1990–2020 and is a standard benchmark in pedagogical implementations of the Vasicek model [5].

Instantaneous volatility $\sigma = 0.02$. With these parameter values, the stationary variance of the process is, from equation (21):

$$\text{Var}_\infty = \frac{\sigma^2}{2\kappa} = \frac{(0.02)^2}{2 \times 0.30} = \frac{4 \times 10^{-4}}{0.60} \approx 6.67 \times 10^{-4} \quad (30)$$

corresponding to a stationary standard deviation of approximately 2.58%. This implies that at stationarity, roughly 95% of simulated rates will lie within $\theta \pm 2 \times 2.58\% = [-0.16\%, 10.16\%]$, which spans a plausible range of interest rate outcomes while keeping negative rates rare—a known limitation of the model discussed formally in Section 7.

Maturity Grid for Yield Curve Construction

Bond prices and yields are computed at $K = 30$ evenly spaced maturities on the grid:

$$\tau_k = \frac{k \cdot T_{\max}}{K}, \quad k = 1, 2, \dots, K \quad (31)$$

with $T_{\max} = 10$ years and $K = 30$, giving a maturity step of $\Delta\tau = 4$ months. This grid spans the conventionally reported segment of the yield curve—from the 4-month bill to the 10-year note—and provides sufficient resolution to distinguish between normal, flat, and inverted curve shapes in the output plots of Section 5.

5 Results and Numerical Analysis

This section presents and interprets the two principal outputs of the computational pipeline described in Section 4: the ensemble of simulated short-rate paths generated by the Euler–Maruyama scheme, and the implied yield curve constructed from the Vasicek affine bond pricing formulae (22)–(25). All results were produced using NumPy’s Mersenne Twister random number generator seeded at `seed = 42`, ensuring full reproducibility.

5.1 Simulated Interest Rate Paths

Visual Output

Figure 1 displays 100 representative sample paths drawn from the full ensemble of $M = 1000$ simulated trajectories of the Vasicek short rate $r(t)$ over

the horizon $t \in [0, 5]$ years. The dashed horizontal line marks the long-run mean $\theta = 0.05$.

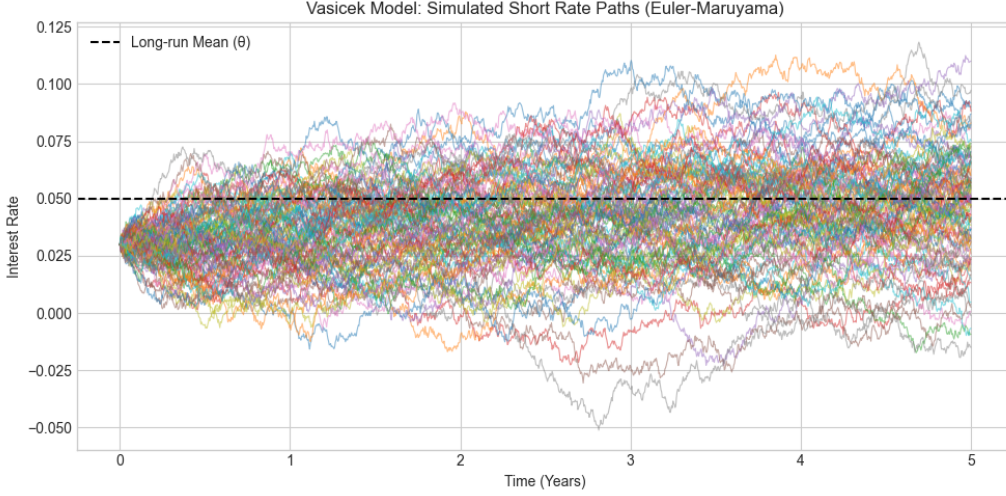


Figure 1: 100 simulated short-rate paths under the Vasicek model, generated via the Euler–Maruyama scheme with parameters $r_0 = 0.03$, $\kappa = 0.30$, $\theta = 0.05$, $\sigma = 0.02$, $\Delta t = 1/252$, and $M = 1000$ total paths. The dashed line indicates the long-run equilibrium θ . Paths were initialized below θ to make the mean-reversion drift visually apparent.

Interpretation: Mean Reversion

The most immediately visible feature of Figure 1 is the upward drift of the path ensemble during the interval $t \in [0, 1]$. This is a direct consequence of the mean-reversion mechanism in the Vasicek drift term $\kappa(\theta - r(t))$: since all paths are initialized at $r_0 = 0.03 < \theta = 0.05$, the drift is strictly positive at $t = 0$, producing a net tendency for rates to rise. Quantitatively, the expected gap between $r(t)$ and θ decays exponentially at rate κ , as established in equation (19):

$$\mathbb{E}[r(t)] - \theta = (r_0 - \theta) e^{-\kappa t} = -0.02 e^{-0.30 t} \quad (32)$$

By $t = 2.31$ years—the half-life computed in equation (29)—the expected deviation has fallen to -0.01 , and by $t = 5$ years it has fallen to $-0.02 e^{-1.5} \approx$

-0.0045 , which is imperceptible on the scale of the figure. This convergence of the ensemble mean to θ is visually confirmed by the centre of mass of the path bundle settling near the dashed line as $t \rightarrow 5$.

Interpretation: Variance Expansion and Stationarity

A second feature is the fan-like widening of the path bundle over time, reflecting the growth of the process variance. From equation (20):

$$\text{Var}[r(t)] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}) = \frac{(0.02)^2}{0.60}(1 - e^{-0.60t}) \quad (33)$$

At $t = 1$: $\text{Var}[r(1)] \approx 6.67 \times 10^{-4}(1 - e^{-0.6}) \approx 3.02 \times 10^{-4}$, corresponding to a standard deviation of $\approx 1.74\%$. By $t = 5$: $\text{Var}[r(5)] \approx 6.67 \times 10^{-4}(1 - e^{-3.0}) \approx 6.17 \times 10^{-4}$, a standard deviation of $\approx 2.48\%$. The variance is therefore approaching—but has not yet reached—its stationary limit of $\sigma^2/(2\kappa) \approx 6.67 \times 10^{-4}$ (standard deviation $\approx 2.58\%$) over the five-year horizon, consistent with the plateau in the spread of the path bundle visible in Figure 1 beyond $t \approx 3$.

Observation: Negative Rate Paths

A subset of paths ventures below zero, most visibly in the range $t \in [2.5, 4]$, with one path reaching approximately $r \approx -0.05$ near $t = 3$. This is not a numerical artifact of the Euler–Maruyama scheme but rather an intrinsic property of the Vasicek model: because $r(t)$ is Gaussian, it assigns strictly positive probability to $r(t) < 0$ for all $t > 0$. At stationarity, the probability of a negative rate is:

$$\mathbb{P}(r > 0) = \Phi\left(\frac{\theta}{\sigma/\sqrt{2\kappa}}\right) = \Phi\left(\frac{0.05}{0.02/\sqrt{0.60}}\right) = \Phi(1.936) \approx 97.4\% \quad (34)$$

so roughly 2.6% of stationary draws are negative. With $M = 1000$ paths and $N = 1260$ time steps, the appearance of isolated negative trajectories in Figure 1 is therefore entirely expected. The structural implications of this feature are discussed in Section 7.

5.2 The Implied Yield Curve

Visual Output

Figure 2 displays the implied yield curve $\tau \mapsto R(0, \tau)$, evaluated analytically at $t = 0$ using the Vasicek affine formulae (22)–(25) at the initial rate $r_0 = 0.03$ over the maturity grid $\tau \in \{1/3, 2/3, 1, \dots, 10\}$ years.

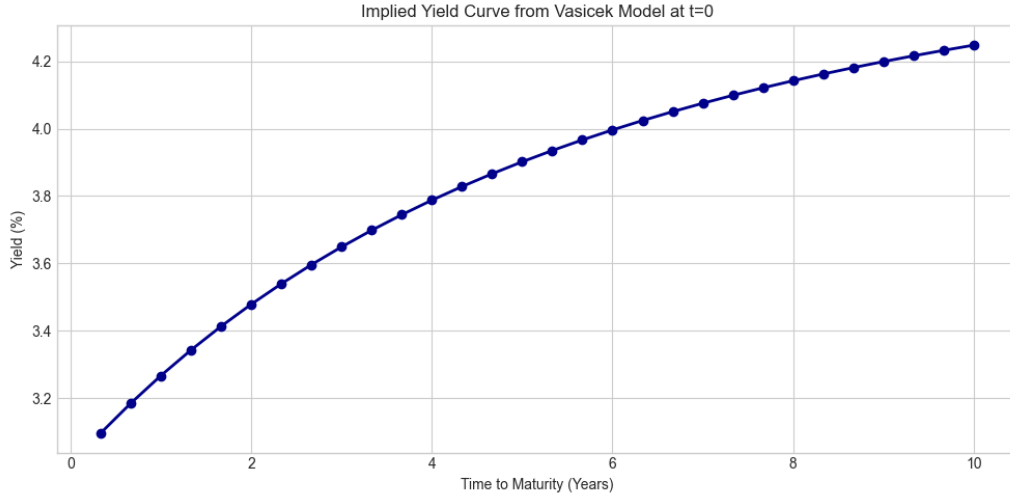


Figure 2: Implied zero-coupon yield curve from the Vasicek model at $t = 0$, computed via the closed-form affine formulae with parameters $r_0 = 0.03$, $\kappa = 0.30$, $\theta = 0.05$, $\sigma = 0.02$. Yields are expressed in percentage terms. The curve is evaluated on a grid of $K = 30$ maturities spanning $[1/3 \text{ yr}, 10 \text{ yr}]$.

Shape: A Normal, Concave Yield Curve

The curve is strictly upward-sloping across all maturities, rising from approximately 3.10% at a 4-month maturity to approximately 4.24% at a 10-year maturity—a total term premium of roughly 114 basis points. This is classified as a **normal** yield curve shape, reflecting the market's expectation that short rates, currently below their long-run mean, will rise over time as the mean-reversion mechanism asserts itself.

The slope is steeply positive at short maturities and flattens progressively toward the 10-year end, producing the characteristic **concave** shape. This

concavity has a precise analytical explanation. From equation (25), the yield $R(0, \tau)$ can be written as:

$$R(0, \tau) = \frac{B(\tau) r_0 - \ln A(\tau)}{\tau} \quad (35)$$

As $\tau \rightarrow \infty$, $B(\tau) \rightarrow 1/\kappa$ and the yield converges to the finite **long-term asymptote**:

$$R_\infty \equiv \lim_{\tau \rightarrow \infty} R(0, \tau) = \theta - \frac{\sigma^2}{2\kappa^2} = 0.05 - \frac{(0.02)^2}{2(0.30)^2} = 0.05 - 0.00\bar{2} \approx 4.78\% \quad (36)$$

The simulated 10-year yield of 4.24% has not yet reached this asymptote, which is consistent with the curve still exhibiting positive slope at $\tau = 10$ years in Figure 2. The asymptote $R_\infty \approx 4.78\%$ lies slightly below the long-run mean $\theta = 5\%$; the gap of approximately 22 basis points is the **convexity correction** $\sigma^2/(2\kappa^2)$, which arises because the nonlinear exponential pricing formula causes bond prices to be convex in r , systematically lowering long-term yields relative to the naive expectation.

Bridging the Two Figures

The two outputs of the simulation are complementary descriptions of the same underlying model. Figure 1 shows the *time-series* behavior of the short rate $r(t)$: its stochastic fluctuations, its mean-reversion to θ , and its eventual stationary distribution. Figure 2 shows the *cross-sectional* behavior at a single instant $t = 0$: how the market prices bonds of increasing maturity given the current short rate $r_0 = 0.03$. The connection between them is the affine bond price formula (22), which integrates the expected future dynamics of $r(t)$ —as characterized by the path ensemble—into a single term structure of discount factors. The normal, concave shape of the yield curve in Figure 2 is therefore a direct reflection of the upward drift and variance expansion visible in Figure 1: because the short rate is expected to rise and its uncertainty grows with maturity, investors demand higher yields for lending over longer horizons.

6 Conclusion

This project has systematically traced the mathematical evolution of interest rate modeling, progressing from the rigid assumptions of deterministic

calculus to the dynamic realism of stochastic differential equations. By dissecting the limitations of first-order ODEs, we established the necessity of incorporating continuous market noise via Brownian motion. This theoretical foundation culminated in the exploration and implementation of the Vasicek model, a canonical framework that mathematically enforces the economic principle of mean reversion through an Ornstein–Uhlenbeck process.

The computational phase of this study successfully bridged continuous-time theory with discrete-time algorithmic application. Utilizing the Euler–Maruyama discretization scheme, we simulated an ensemble of short-rate trajectories. As demonstrated in Section 5.1, these simulated paths visually corroborated the core theoretical properties of the SDE: an exponential decay toward the long-run mean and a bounded variance expansion approaching stationarity.

Furthermore, the application of affine term structure formulae demonstrated the fundamental link between time-series rate dynamics and cross-sectional bond pricing. By translating our simulated initial state into an implied yield curve in Section 5.2, we generated a financially intuitive normal, concave term structure. The mathematical emergence of the convexity correction in our long-term asymptote highlights the non-linear realities of fixed-income pricing that simpler, deterministic models inherently fail to capture.

Ultimately, this study underscores the indispensable role of stochastic calculus and numerical simulation in modern quantitative finance. By synthesizing theoretical derivations, computational methods, and financial interpretation, we have constructed a robust and highly transparent methodological pipeline for yield curve analysis.

7 Limitations

While the Vasicek model provides an elegant, closed-form framework for understanding mean reversion and term structure dynamics, its assumption of constant volatility leads to a fundamental structural limitation: it permits strictly negative interest rates.

Because the stochastic driver is a standard Brownian motion scaled by a constant σ , the short rate $r(t)$ follows a Gaussian distribution for all $t > 0$. As calculated in equation (34) and observed in the simulated paths of Figure 1, the process can—and almost surely will, given a sufficiently long horizon—

cross below zero. In classical macroeconomic environments, this violates the basic axiom of the time value of money, as investors would historically prefer to hold physical cash rather than accept negative nominal yields. Although some central banks have experimented with sub-zero policy rates in recent years, a model that systematically generates deeply negative rates at stationarity remains highly problematic for long-term derivative pricing and risk management.

8 Future Work: The Cox–Ingersoll–Ross (CIR) Model

The immediate direction for future research involves resolving the negative rate boundary by transitioning to a volatility specification that depends on the state variable. The canonical extension is the **Cox–Ingersoll–Ross (CIR) model**, which defines the short-rate SDE as:

$$dr(t) = \kappa(\theta - r(t))dt + \sigma\sqrt{r(t)}dW(t) \quad (37)$$

By introducing the square-root term $\sqrt{r(t)}$ into the diffusion coefficient, the CIR model ensures that as the interest rate approaches zero, the instantaneous volatility also approaches zero. This effectively dampens the downward random shocks near the origin. Furthermore, if the parameters satisfy the **Feller condition**:

$$2\kappa\theta \geq \sigma^2 \quad (38)$$

the deterministic upward drift at the boundary is sufficiently strong to guarantee that the rate $r(t)$ remains strictly positive for all t .

Implementing the CIR model computationally introduces new numerical challenges, as the standard Euler–Maruyama scheme can produce negative intermediate values that render the square root undefined. Future work would require implementing specialized discretization techniques, such as a truncated Euler scheme, to simulate the CIR process and compare its resulting yield curve dynamics against the Vasicek baseline established in this study.

A Python Implementation

The following Python script was used to execute the Euler–Maruyama simulation and generate the figures presented in Section 5.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # 1. Parameter setup (refer to Section 3.2)
5 r0 = 0.03      # initial rate = 3%
6 kappa = 0.30   # mean reversion speed
7 theta = 0.05   # long-run mean
8 sigma = 0.02   # volatility
9 T = 5.0        # 5 year horizon
10 N = 1260       # 252 trading days * 5
11 dt = T / N
12 M = 1000       # total simulated paths
13
14 # 2. Random noise generation
15 # setting seed so the plots in the report don't change on re-
16   run
17 np.random.seed(42)
18 # drawing all standard normals at once (much faster than
19   looping)
20 Z = np.random.standard_normal((M, N))
21
22 # 3. Euler-Maruyama Simulation
23 r = np.zeros((M, N + 1))
24 r[:, 0] = r0
25
26 for i in range(N):
27     # updating all M paths simultaneously to save compute
28     # time
29     drift = kappa * (theta - r[:, i]) * dt
30     diffusion = sigma * np.sqrt(dt) * Z[:, i]
31     r[:, i + 1] = r[:, i] + drift + diffusion
32
33 # 4. Implied Yield Curve (Vasicek Affine Formula)
34 T_max = 10.0
35 K = 30
36 tau = np.linspace(T_max / K, T_max, K)
37
38 yield_curve = np.zeros(K)
39
40 for k, t in enumerate(tau):
```

```

38     # A and B factors
39     B = (1 - np.exp(-kappa * t)) / kappa
40     A = np.exp((theta - (sigma ** 2) / (2 * kappa ** 2)) * (B
41         - t) - (sigma ** 2 / (4 * kappa)) * B ** 2)
42
43     # price the zero-coupon bond given current r0
44     P = A * np.exp(-B * r0)
45
46     # compute continuously compounded yield
47     yield_curve[k] = -np.log(P) / t
48
49 # 5. Plotting results
50 plt.style.use('seaborn-v0_8-whitegrid')
51
52 # Figure 1: Simulated Paths
53 plt.figure(figsize=(10, 5))
54 time_grid = np.linspace(0, T, N + 1)
55
56 # only plotting the first 100 paths so the graph isn't a
57 # solid block of color
58 for j in range(100):
59     plt.plot(time_grid, r[j, :], lw=0.8, alpha=0.5)
60
61 plt.axhline(theta, color='black', linestyle='--', label='Long
62     -run Mean (theta)')
63 plt.title('Vasicek Model: Simulated Short Rate Paths (Euler-
64     Maruyama)')
65 plt.xlabel('Time (Years)')
66 plt.ylabel('Interest Rate')
67 plt.legend()
68 plt.tight_layout()
69 plt.savefig('plot1.jpg', dpi=300, bbox_inches='tight')
70 plt.show()
71
72 # Figure 2: Yield Curve
73 plt.figure(figsize=(10, 5))
74 plt.plot(tau, yield_curve * 100, color='darkblue', marker='o',
75     , linestyle='-', linewidth=2)
76 plt.title('Implied Yield Curve from Vasicek Model at t=0')
77 plt.xlabel('Time to Maturity (Years)')
78 plt.ylabel('Yield (%)')
79 plt.tight_layout()
80 plt.savefig('plot2.png', dpi=300, bbox_inches='tight')
81 plt.show()

```

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